

Entity-Level Sentiment from Financial News

Model, training, validation, and tests against S&P 500 returns and volatility

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Abstract

This report summarises a two-part project. The first part is a model that reads a complete financial news article and assigns every company, ticker, person and organisation it finds a sentiment score in $(-0.95, 0.95)$ that expresses the *implication of the news for that entity*, not the tone of the article. It is a shared Longformer-large encoder (2,048 tokens per article) with a CRF-decoded entity tagger and a cross-attention sentiment head, 439M parameters in total, trained by distillation from LLM-generated labels: three stages on about 20,000 articles, then a head-only retrain on about 30,000 relabelled articles with a concordance-based loss. On a held-out test split the sentiment scores reach a Pearson correlation of 0.64 with the labels and match their spread; the entity tagger, which has to find the entities before they can be scored, is the bottleneck at about 48% coverage. The second part scores 118,033 wire-grade articles about 502 S&P 500 constituents (2020–2025) and tests whether the scores predict returns. They do not: the apparently strong first-pass evidence (one-day rank IC +0.020, Newey–West t 3.7; a close-entry long-short Sharpe of 2.9) was look-ahead, because about a quarter of the articles, including the after-hours earnings spike, had been dated to a session whose close preceded their publication. Rebuilt with a 15:45 ET knowability cutoff, the IC is indistinguishable from zero at every horizon and every daily-bar execution tested loses money. Two results survive: the scores align strongly with the contemporaneous overnight reaction, which makes the model a useful measurement instrument, and the *magnitude* of sentiment predicts next-day absolute returns beyond trailing realised volatility (rank IC +0.024, t 5.3).

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*Independent project, built on the author’s own time, hardware and data subscription; no employer code or data is involved. The code and analyses were produced with Claude Code under the author’s direction and were reviewed read-only by two other model-based reviewers (Codex and Kimi); all conclusions were verified by the author. Code and this report: https://github.com/Russell-Ding/entity_sentiment_model_pipeline.

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1 Summary of findings

The project asked five questions. Table 1 gives the short answers; the rest of the report gives the evidence. Sections 2–5 cover the model and Section 6 the market tests; Appendix A reproduces the labelling prompts and Appendix B lists what the public repository contains.

Question	Answer	Evidence
Can a long-document model score sentiment per entity?	Yes, moderately well	v2.1 held-out test: Pearson 0.64, concordance 0.64, prediction spread equal to the labels'; 95% of entities within one sentiment bucket of the label; strongest on tickers (r 0.81).
Does the end-to-end pipeline find the entities?	Only about half	Entity coverage 42–48%; span precision 0.86 but recall 0.47 on sentiment-bearing types.
Do the scores predict next-day returns?	No detected edge	Knowability-corrected one-day rank IC -0.007 (t -1.4); the implementable market-on-close strategy has gross Sharpe -0.55 ; next-open entry is negative at every holding period tested. One channel, pre-market news at the same day's open, remains untested.
Do the scores explain the market's reaction?	Yes	First-pass day-one quintile spread $+36$ bps, of which $+41$ bps is the overnight gap after publication and -5 bps the tradable intraday part.
Does sentiment predict volatility?	Yes, modestly	$ \text{sentiment} $ vs. next-day $ \text{return} $ after trailing realised volatility: rank IC $+0.024$ (t 5.3); $+0.019$ (t 3.7) with an extra same-day $ \text{return} $ control (independent replication). Not yet hardened or reproducible from a committed script.

Table 1: The project at a glance. IC = information coefficient (daily cross-sectional Spearman correlation), t = Newey–West t -statistic, bps = basis points.

Two provenance notes. First, the market tests were run on scores produced by the v2.0 checkpoint, not v2.1 as the study's working documents state: the inference files predate the v2.1 release, the inference launcher hard-codes the v2.0 checkpoint, and re-scoring stored articles reproduces the saved values only with v2.0. The corrected attribution is used throughout this report, and a correction note has been added to the study documents. Second, everything here was built with model-based tooling; the first-pass conclusion of the market study was wrong, and the correction is reported in full because the correction is the substantive result.

2 Data and labels

2.1 News corpus

The corpus is the EODHD news feed for the S&P 500 universe: 1.69 million articles from 2020 to 2025, stored per primary ticker with a publication timestamp, source and URL. Every article was assigned a *source tier* and a content type by a rule set distilled from a 5,000-article LLM-labelled sample: tier 1 is wire-grade reporting (Reuters, Bloomberg, AP, WSJ and similar, 9.6% of the corpus), tier 2 is secondary and analytical press including market wraps, analyst-call pieces and earnings transcripts (9.0%), and tier 3 is algorithmic and search-optimised bulletins, syndicated press releases and listicles (81.3%). The tiers were designed to carry a per-article loss weight; in the released training path they select which articles enter the training subsets rather than reweighting the loss. The market study used tier-1 articles only.

Coverage is uneven, and this matters downstream. The feed carries no Bloomberg or AP content for November–December 2024 (a provider-side outage, confirmed by inspecting the raw feed), and it has no tier-1 coverage of NVIDIA’s earnings week of 23–29 May 2023. A coverage-gap flag was added to the daily series so that readings following a blackout can be identified. One wire service contributes about 14,500 tier-1 items that turned out to be 300-character paywall stubs; they were marked for demotion to tier 2 but the corpus was never re-tiered, so they are filtered by the per-ticker daily aggregation and *not* by the 502-name signal panel of Section 6, which reads the inference files directly. The news and price data are not redistributed (EODHD terms).

2.2 Labels by distillation

No human-annotated corpus exists for entity-level financial sentiment at this granularity, so labels were generated by large language models from a written specification and then filtered. Three label sets were produced over the life of the project (Table 2); the model versions in Section 4 map onto them one to one.

Label set	Labeller	Size and splits	Used for
v1 (Jan 2026)	Claude Haiku 4.5	2,133 train / 235 validation articles; COMPANY entities only	v1.0 model (Longformer-base)
Sonnet relabels (Feb–Apr 2026)	Claude Sonnet, simplified prompt	17,587 train / 2,130 validation; evaluation holdouts of 6,750 relabelled and 9,371 original-label articles, drawn from non-overlapping sources	v2.0 model, all three training stages
Decisive set (Jun 2026)	DeepSeek v4-flash with the <i>decisive</i> prompt; PERSON entities re-scored by Kimi	30,131 tier-1 articles after near-duplicate removal, split by article id into 27,015 / 1,558 / 1,558 (train / validation / test)	v2.1 head retrain and its in-distribution evaluation; the Sonnet holdout is kept for the tagging and cross-distribution checks

Table 2: Label sets. The two prompts used for the decisive set are reproduced in Appendix A.

Each labelled article carries a list of entity records. A record has a canonical id and name, a type

(ORG, TICKER, PERSON, MONEY, PERCENT, DATE), the literal name mentions in the text, the coreferent mentions (*it*, *the company*, *the iPhone maker*) kept separately, and a sentiment score in $[-1, 1]$ for ORG, TICKER and PERSON records; MONEY, PERCENT and DATE records are extracted but carry no score. A company and its ticker symbol, when both appear literally, are separate records that share one score. Mentions are given as strings and located in the text afterwards by code, which removed the character-offset errors that plagued the first label set.

2.3 What the score means

The scoring rule is the part of the specification that changed most between label sets, and it is the reason v2.1 exists. The Sonnet prompt asked for the *tone* of the article with respect to the entity, and 0.0 for “factual mention”. Financial news is written in a flat, factual register, so that rule produced a label distribution heavy in zeros and a model that hedged toward zero. The decisive prompt scores the *implication* of the news for the entity even when it is reported neutrally (a lawsuit defendant is negative; an earnings beat is positive) and reserves 0.0 for entities with no stake in the story (an analyst’s employer cited for a rating, a data source, an exchange). Table 3 gives the bands. About 59% of the scored entities in the decisive set are still incidental, which shapes the loss design in Section 4.2.

Score	Meaning	Typical trigger
−1.0 to −0.7	very negative	bankruptcy, fraud, criminal charges
−0.6 to −0.4	moderately negative	lawsuit or antitrust defendant, earnings miss, downgrade, layoffs, recall, probe
−0.3 to −0.1	slightly negative	cautious outlook, minor headwind, lagging a rival
0.0	neutral	incidental, no-stake mention only
+0.1 to +0.3	slightly positive	minor beat, modest gain, favourable mention
+0.4 to +0.6	moderately positive	earnings beat, upgrade, major deal or launch, winning a suit
+0.7 to +1.0	very positive	record results, blockbuster product, transformative win

Table 3: Sentiment bands in the decisive labelling prompt (Appendix A).

2.4 Quality controls

Near-duplicate articles (wire re-runs with distinct ids) were removed before splitting, and the splits are disjoint by article id. An automated quality guardian sampled the completed labels during each run and halted it when format, extraction, neutrality or failure rates breached a threshold.

PERSON scores from the first DeepSeek pass were noticeably worse than ORG scores, so PERSON entities were re-scored with a dedicated prompt wherever the request was accepted; cases the re-scoring model declined kept their original score. Measured afterwards with the *unchanged* v2.0 model, the re-scored entities correlate with their labels at 0.50 and the 1,334 that were not re-scored at 0.29. These are different subsets rather than a before-and-after measurement of the same cases, so the comparison suggests that the re-score fixed labels rather than fitting the model, without establishing it.

After the v2.1 retrain, the highest-confidence disagreements between model and label were extracted for manual adjudication. The v2.1 model card records the outcome as about one clear label error in roughly 24 adjudicated cases, which is the basis for treating the retrain gains as real rather

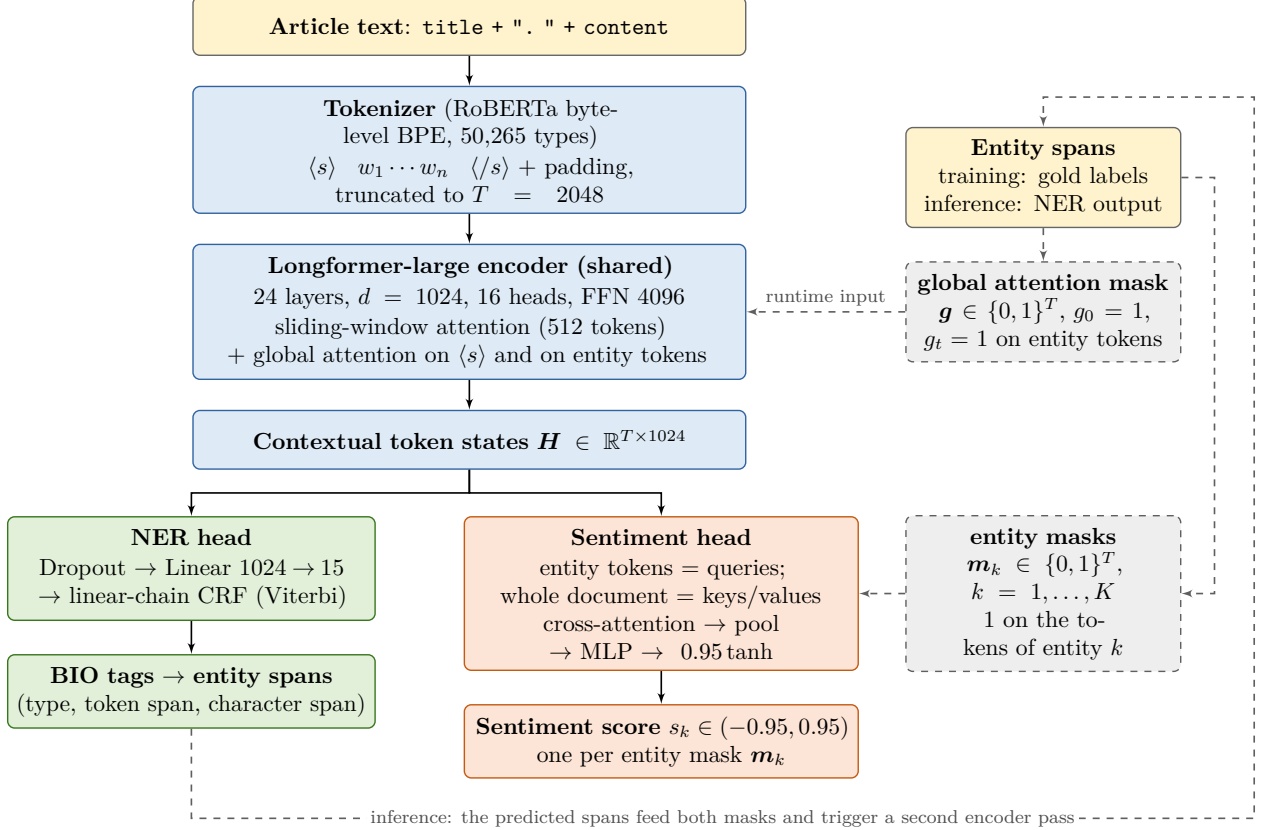


Figure 1: Model structure. Solid arrows are the forward tensor flow; dashed arrows carry the two masks through which entity positions enter the network, and the inference feedback path that rebuilds those masks from the predicted spans.

than fitted noise; the released audit file lists the cases and the disagreement rates but not the adjudications themselves.

3 Model

Figure 1 shows the model. One shared encoder produces contextual token states; two small heads read them. The heads are coupled through two masks that carry entity positions: the *entity masks* select which token states become the queries of the sentiment head, and the *global-attention mask* tells the encoder itself which tokens may attend to the whole document. During training the masks come from the gold entity mentions; at inference they come from the tagger’s predicted spans, so the encoder is run twice per article.

Encoder. allenai/longformer-large-4096 [1]: 24 layers, hidden size 1,024, 16 heads, a 512-token sliding attention window plus global attention on selected tokens. Articles are truncated or padded to exactly 2,048 tokens, half of the encoder’s 4,096-position capacity; the budget was chosen for GPU memory and throughput, and longer articles lose their tail. Longformer was chosen over 512-token encoders because the sentence that settles an entity’s sentiment is often far from its mention, and the global-attention mechanism gives a natural way to make the encoder *entity-aware*: every token can attend to the entity tokens and the entity tokens can attend to the whole article.

NER head. Dropout, a linear layer to 15 BIO tags (*O* plus begin/inside tags for COMPANY, TICKER, PERSON, ORG, MONEY, PERCENT and DATE; the COMPANY tag is a legacy of the first label set and unused by v2 labels, which file companies under ORG) and a linear-chain CRF decoded by Viterbi. Tags are converted to spans with a lenient decoder that repairs an *inside* tag without a preceding *begin*. In the released v2 weights the CRF transition scores never moved from their initial values (Section 4.4), so the tagger behaves as a per-token classifier followed by span repair.

Sentiment head. For each entity mask, the entity’s token states become queries and the whole document’s states become keys and values of an 8-head cross-attention (with layer normalisation on both sides and a residual connection); the attended entity tokens are mean-pooled, normalised, and mapped by a three-layer MLP ($1024 \rightarrow 512 \rightarrow 128 \rightarrow 1$, GELU, dropout) to a scalar that is squashed as $s = 0.95 \tanh(z)$. The head has 4.80M parameters. All entities of an article are scored in one call, so an article with K entities costs one encoder pass plus one batched head evaluation.

Inference: two encoder passes. Pass 1 runs the encoder with global attention on the $\langle s \rangle$ token only and decodes entity spans. Pass 2 runs it again with global attention on $\langle s \rangle$ plus every predicted sentiment-bearing span and scores each span. Spans with the same surface text and type are averaged into one record per entity per article; MONEY, PERCENT and DATE spans are dropped. Making a single encoder serve both regimes is what the training procedure has to solve (Section 4).

Component	Parameters	Trained in
Longformer-large encoder	434,600,960	Stages 1–2 (frozen afterwards)
NER head (linear + CRF)	15,630	Stages 1–2 (frozen afterwards)
Sentiment head (cross-attention + MLP)	4,795,137	Stage 3; retrained for v2.1
Total	439,411,727	

Table 4: Parameter budget of the v2.x model (read from the checkpoint).

4 Training procedure

4.1 Why the recipe has stages

An earlier version of the model was trained end to end with global attention always placed on the gold entity tokens. With gold entity positions it tagged well (F1 0.74), but in real use there are no gold positions: the tagger must run with global attention on $\langle s \rangle$ alone, a regime the encoder had never seen, and end-to-end F1 collapsed to 0.05. The fix, *global-attention dropout*, hides the entity-aware global attention for a random 30% of training samples so that the encoder learns to produce useful states in both regimes. Around that idea the recipe became three stages that train progressively fewer parameters, followed by a head-only retrain when the labels changed (Table 5).

Two details of the sequence are easy to miss. Stage 3 loads the *Stage-1* checkpoint, not Stage 2’s, because Stage 3 discards whatever sentiment head it inherits and only needs the best encoder for the CLS-only tagging regime. And the v2.1 head was *warm-started* from the v2.0 head rather than re-initialised: the retrain script loads the full v2.0 state and never resets the head, and the run ledger’s pre-training baseline for the sibling arm is the v2.0 head’s own validation score. (An earlier version of the v2.1 model card described the head as re-initialised; the card’s corrections section now fixes this.)

Stage	Trains	Data	Loss and settings	Outcome
1. NER	encoder + NER head (all parameters)	Sonnet relabels: 17,587 train / 2,130 val	CRF negative log-likelihood only (sentiment weight 0). 5 epochs, batch 24, AdamW, LR 2×10^{-5} , mixed precision, gradient checkpointing; global-attention dropout $p = 0.3$, ramped up over the first 30% of steps	Best CLS-only NER F1 0.584 at epoch 2 (entity-aware F1 about 0.76); this checkpoint feeds Stage 3
2. Joint	everything	same	CRF loss + sentiment loss with a curriculum: NER weight $1.0 \rightarrow 0.5$, sentiment weight $0.3 \rightarrow 1.0$. 5 epochs, batch 24, LR 10^{-5} , dropout $p = 0.3$ without ramp	Joint training pulled the encoder away from the CLS-only optimum (CLS-only F1 to 0.53); its checkpoint was therefore not used downstream
3. Sentiment	sentiment head only (freshly initialised); encoder and NER head frozen, loaded from the Stage-1 checkpoint	same	$0.5 \text{ MSE} + 0.5 (1 - \rho)$ with ρ the batch Pearson correlation (Eq. 1). 10 epochs, batch 80, LR 5×10^{-4} , no attention dropout	Best epoch 9: validation r 0.511. Packaged as v2.0 (17 May 2026)
v2.1 head retrain	sentiment head only, initialised from the v2.0 head; encoder and NER head frozen	Decisive set: 27,015 / 1,558 / 1,558	weighted Huber + $(1 - \text{CCC})$ + sign penalty (Eq. 2). 10 epochs, batch 48, head LR 5×10^{-4} , cosine schedule, best epoch by validation CCC	Best epoch 10: validation CCC 0.618. Packaged as v2.1 (20 June 2026); encoder and NER head byte-identical to v2.0

Table 5: The training sequence. The documented Colab recipe uses a single high-memory GPU and estimates roughly 14 hours for Stages 1–3 together; per-run hardware and timing logs are not part of the release.

4.2 Loss functions

Sentiment is a regression on the scored entities of a batch. The v2.0 head used

$$L_{\text{v2.0}} = \frac{1}{2} \text{MSE}(\hat{y}, y) + \frac{1}{2} (1 - \rho(\hat{y}, y)), \quad (1)$$

which has a specific weakness. On a label distribution that is 59% zeros, mean-squared error is minimised by shrinking uncertain predictions toward zero, and the Pearson term is scale-invariant, so it cannot see the shrinkage: a model that predicts exactly half the true magnitude has $\rho = 1$. The trained v2.0 head accordingly predicted with 57% of the labels’ standard deviation and almost never emitted a “very negative” score.

The v2.1 loss replaces both terms:

$$L_{\text{v2.1}} = \frac{\sum_i w_i \text{Huber}_\delta(\hat{y}_i - y_i)}{\sum_i w_i} + (1 - \text{CCC}(\hat{y}, y)) + 0.5 \cdot \text{mean}_{|y_i| \geq 0.4} \text{ReLU}(-\hat{y}_i y_i), \quad (2)$$

$$w_i = \min(1 + 3|y_i|, 3), \quad \delta = 0.1,$$

where the concordance correlation coefficient [2]

$$\text{CCC} = \frac{2 \text{cov}(\hat{y}, y)}{\text{var}(\hat{y}) + \text{var}(y) + (\mu_{\hat{y}} - \mu_y)^2} \quad (3)$$

measures agreement with the identity line rather than with any line, so magnitude compression and mean shift are penalised directly. The Huber term (a weighted mean, normalised by the sum of the weights) is robust to the noisiest labels, which are the LLM’s extremes; the magnitude weights stop the neutral majority from driving the fit to zero; and the hinge term adds an explicit penalty when a strongly scored entity is predicted with the wrong sign. Model selection also changed, from validation Pearson correlation to CCC: a scale-invariant selection metric cannot see magnitude compression either.

4.3 The v2.1 retrain and its ablation

Evaluating v2.0 on the new decisive labels (Section 5) showed a correlation of 0.485 and the compressed spread noted above. Before retraining, a cheap alternative was tested: fitting an affine or isotonic calibration to v2.0’s own predictions. The optimal affine slope came out at 0.86, below one, and left the correlation unchanged, which established that the compression was regression attenuation (the correct hedge for a model with limited discrimination), not a fixable slope. A five-arm ablation harness was then defined over the same v2.0 checkpoint and splits: the old loss on the new labels, the new loss (the *control* arm), the new loss with the top two encoder layers unfrozen (*arm C*), and two loss-component ablations. Training results exist for two of them, the control and arm C, and those are the arms compared in Table 6.

Model	N	Pearson	CCC	std ratio	sign flips	F1 v-neg	F1 v-pos
v2.0 baseline (full new set)	164,723	0.485	0.416	0.57	0.215	0.00	0.01
v2.0 + affine calibration	82,628	0.483	–	0.49	0.204	0.00	0.00
control (= v2.1): new loss, frozen encoder	8,078	0.643	0.641	1.03	0.151	0.13	0.37
arm C: new loss, top-2 layers unfrozen	7,780	0.690	0.686	1.10	0.134	0.15	0.38

Table 6: Retrain ablation on the decisive labels. “std ratio” is the standard deviation of the predictions over that of the labels; “sign flips” is the share of entities with both label and prediction away from zero whose signs disagree. The v2.0 baseline is measured on the full new-label set (which contains the test split) and the calibration row on the held-out half of that set; the two arms are measured on the 1,558-article test split. No v2.0 baseline was run on the test split alone, so the arm-versus-baseline gaps are not a same-split comparison.

Arm C scored higher in distribution but was rejected: unfreezing the encoder cost 6.8 points of PERSON tagging F1 on the gold benchmark (catastrophic forgetting in the shared encoder) and shifted the calibration on the cross-distribution holdout for no ranking gain. The control arm became v2.1, so the v2.x encoder and tagger are unchanged since May 2026 and only the sentiment head differs between v2.0 and v2.1.

4.4 A training defect discovered after release

While documenting the model, an inspection of the released weights showed that all 225 CRF transition scores and the 30 start/end scores lie within ± 0.11 , barely beyond the ± 0.1 range of the library’s random initialisation, and that the BIO-validity constraints the head can install were never present: the training script built the head without the label map that installs them, and the constraints written at load time are overwritten by the checkpoint. Emission scores are larger by a factor of roughly 35 to 65, and in a spot check of three validation articles Viterbi decoding agreed with per-token argmax on more than 99% of tokens. The tagger is therefore effectively

emission-only; span validity comes from the lenient decoder. This does not change any number reported here (all evaluations used the released weights), but a future retrain should pass the label map at construction and give the CRF parameters their own learning rate. A script that reproduces the measurement is in the repository (`scripts/analysis/inspect_ner_head.py`).

5 Sentiment prediction quality

5.1 Evaluation protocol

Two protocols are used. The *gold-mask* protocol gives the model the labelled entity positions and measures the sentiment head alone. The *end-to-end* protocol, which is the one that matters for use, starts from raw text: pass 1 tagging, pass 2 scoring, then predicted spans are matched to gold mentions by character overlap ($\text{IoU} \geq 0.5$), the predictions matched to one gold entity are averaged, and three things are reported: span-level tagging precision, recall and F1; *coverage*, the share of labelled sentiment-bearing entities for which the pipeline produced any score; and the sentiment metrics on the covered entities. Sentiment metrics on covered entities are optimistic by a few points, because the entities the tagger finds tend to be the easier ones.

5.2 v2.0 on the Sonnet holdout

Table 7 reports v2.0 on the 6,750-article relabelled holdout, which is drawn from sources not used in training. The tagger is conservative (precision 0.71, recall 0.34); the sentiment correlation on covered entities is 0.63, highest for tickers and lowest for people.

v2.0, end to end, Sonnet holdout (6,750 articles)	
NER span F1, sentiment-bearing types (precision / recall)	0.459 (0.713 / 0.338)
Entity coverage	42.2% (27,566 of 65,270)
Sentiment Pearson r on covered entities	0.628
by type: TICKER / ORG / PERSON	0.773 / 0.598 / 0.412
Mean absolute error	0.192
Exact / adjacent sentiment-bucket accuracy (5 buckets)	0.564 / 0.948
Gold-mask holdout (9,371 articles): sentiment r / NER F1	0.525 / 0.737

Table 7: v2.0 evaluation. The gold-mask NER F1 is measured with entity-aware global attention, the end-to-end F1 with CLS-only attention, which is why they differ so much.

5.3 v2.0 versus v2.1 on the decisive labels

Table 8 is the comparison that motivated and then validated the retrain. With the same encoder and tagger (coverage 48%), the new head raises the correlation from 0.49 to 0.64, restores the spread of the predictions (std ratio 0.57 to 1.03), cuts the rate of flat predictions on strongly scored entities by about a third, reduces sign flips, and for the first time produces non-trivial “very negative” and “very positive” predictions. The two columns come from different evaluations rather than one held-out comparison, so they bound the improvement rather than measure it exactly. Bootstrap

confidence intervals are grouped by article, since entities within an article are not independent; they are quoted from the model card, as the bootstrap output itself is not among the released artefacts.

Decisive labels	v2.0 (full set, 164,723)	v2.1 (test split, 8,078)
Pearson r	0.485	0.643 [0.626, 0.660]
Spearman ρ	0.451	0.600
CCC	0.416	0.641
Prediction / label std ratio	0.57	1.03
Mean absolute error	0.209	0.196
Flat prediction ($ \hat{y} < 0.1$) when $ y \geq 0.4$	30.8%	21.3%
Sign-flip rate	21.5%	15.1%
Very-negative / very-positive F1	0.00 / 0.01	0.13 / 0.37
Pearson r by type: ORG / TICKER / PERSON	0.486 / 0.622 / 0.493	0.627 / 0.807 / 0.678
Exact / adjacent bucket accuracy	0.562 / 0.955	0.582 / 0.953
Coverage (same tagger)	48.9%	48.4%
NER span F1, sentiment-bearing types (P / R)	0.603 (0.856 / 0.465)	0.610 (0.857 / 0.473)

Table 8: Effect of the v2.1 head retrain. The v2.0 column is measured on the whole decisive set (which includes the test split); the v2.1 column on the held-out test split only. The TICKER correlation for v2.1 rests on 112 entities and should be read as indicative.

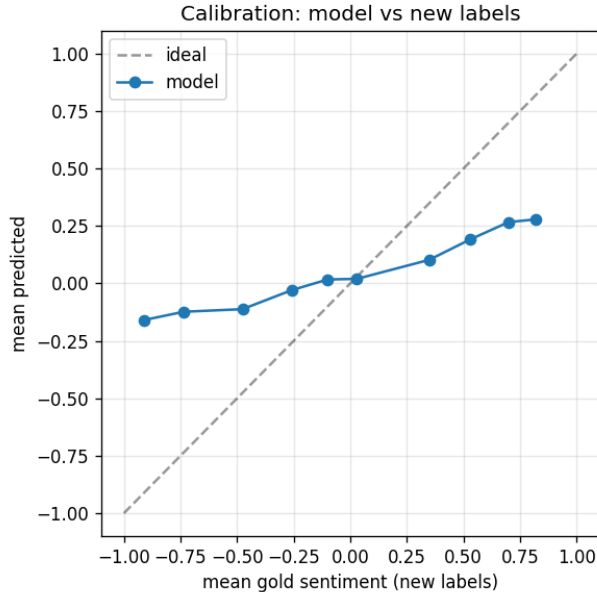


Figure 2: Calibration of the v2.0 head against the decisive labels: mean prediction per label decile. The slope well below one is the under-polarisation that the v2.1 loss was designed to remove; on the v2.1 test split the prediction spread equals the label spread.

Cross-distribution check. On the Sonnet holdout, which follows the older tone-based scoring rule, v2.1 keeps most of v2.0’s correlation (0.606 vs. 0.628), keeps a calibrated spread (std ratio 1.02 vs. 0.73) and acquires a mild positive mean bias of +0.06. The head has learned the decisive rule rather than overfitting the DeepSeek labeller’s style, and a per-deployment affine offset would remove the bias if absolute scores were consumed on a tone-labelled corpus.

5.4 Tagging

Table 9 breaks the v2.1 tagger down by type on the test split. Precision is high for organisations and tickers and recall is low everywhere; dates and percentages are barely recalled, but they carry no sentiment. Labelling a ticker only when the symbol appears literally makes TICKER spans rare, so that row rests on 1,127 gold spans and should be read as an estimate with wide error bars; it does not by itself explain why recall on those spans is low.

Type (v2.1, test split)	Precision	Recall	F1	Gold spans
ORG	0.885	0.564	0.689	34,464
PERSON	0.694	0.231	0.347	11,372
TICKER	0.861	0.148	0.253	1,127
MONEY	0.802	0.352	0.489	5,071
PERCENT	0.670	0.176	0.278	5,677
DATE	0.863	0.061	0.114	11,879
All types	0.844	0.370	0.514	69,590

Table 9: Span-level tagging by entity type, end to end with CLS-only global attention.

5.5 Error modes

The audit of high-confidence disagreements characterised the residual errors as model behaviours rather than label noise:

- **Legal outcomes.** The model anchors on negative keywords (*ban*, *sue*, *probe*, *antitrust*) even when the entity *won*: “Apple wins bid to pause Watch ban” scored -0.51 against a label of $+0.5$.
- **Secondary and list-style mentions** in market wraps are under-read. About 3.7% of all covered entities carry a strong label but receive a flat score; expressed as a share of the strongly labelled entities alone the rate is 21.3% (Table 8).
- **Charged context around an incidental entity** (a company named only as a benchmark or partner) is over-read.
- **Coverage.** Sentiment is produced only for the roughly half of entities the tagger finds; improving recall is the largest available gain and would also change the composition of every downstream aggregate.

5.6 The scored corpus, illustrated

Figure 3 shows one ticker’s monthly sentiment from the corpus scored for the market study. The broad shape (a trough in the 2022 GPU downturn, a rise through the 2023–24 AI build-out, softening in 2025) is plausible, but two features are data artefacts rather than model outputs: the May 2023 dip sits on a week with no tier-1 coverage at all, and the sharp rise in article counts in 2025 partly reflects a new wire source entering the tier-1 set. Monthly aggregates at this granularity are noisy; the reviewers’ recommendation to read them at daily granularity around known events stands.

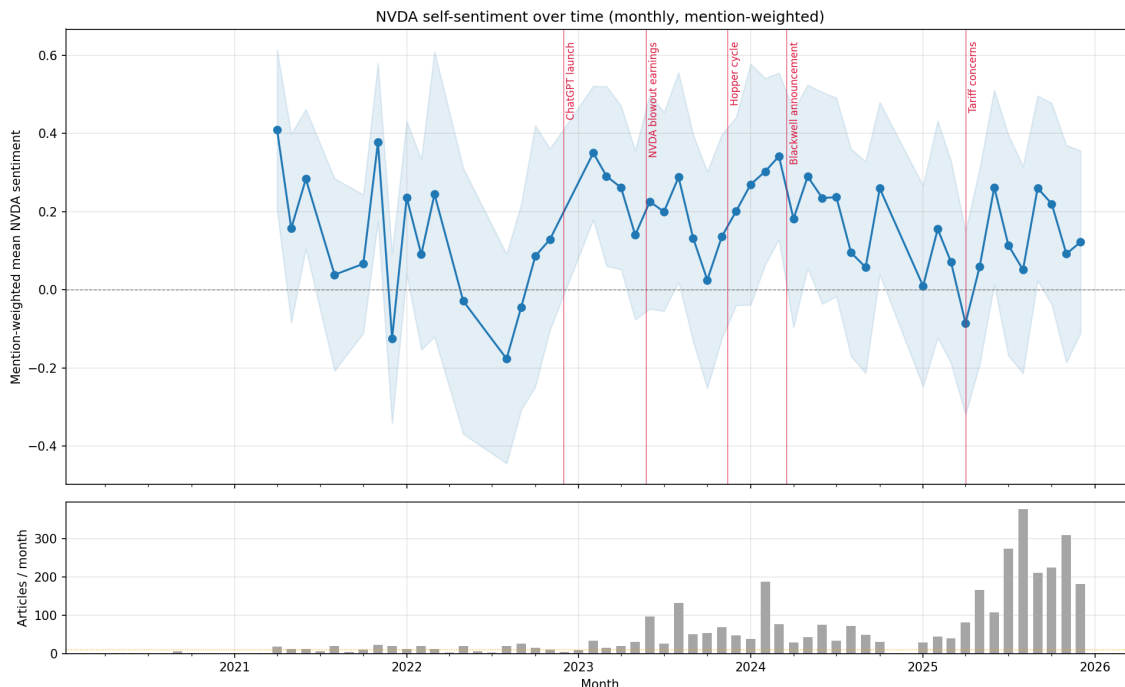


Figure 3: Monthly mention-weighted NVIDIA sentiment (v2.0 scores over tier-1 articles) with article counts. Months with fewer than ten articles are omitted.

6 Do the scores predict returns or volatility?

6.1 From article scores to a daily panel

All tier-1 articles for the S&P 500 universe were scored end to end with the v2.0 checkpoint (118,033 unique articles, May–June 2026 inference run). Each article’s entity records were resolved to tickers with a constituent list plus a hand-made alias table and a blocklist of non-companies (regulators, exchanges, news outlets), and the per-ticker score of an article is the mention-weighted mean over the aliases that resolved to it. The daily signal `sent_mw` for a (ticker, date) pair is the mention-weighted mean over that day’s articles. The date attached to each article at inference time was the calendar date of its publication timestamp, taken as written and with no conversion to New York time; articles whose date had no trading session, such as weekend and holiday items, had no price bar to join to and were dropped from the panel. This dating rule is what Section 6.4 shows to be wrong. The vendor-polarity baseline used a different rule, rolling each article forward to the next available session.

The panel joins the signal to EODHD daily prices (open, close, adjusted close) and attaches forward close-to-close returns at 1, 5 and 20 trading days, a size proxy (log dollar volume) and 1- and 12-month momentum controls. It has 60,888 ticker-days across 502 tickers from 23 March 2020 to 31 December 2025 (1,441 trading days) with a median of 45 names per day (10th–90th percentile 8 to 64). The panel keeps thin days; each test applies its own floor, five usable names for the information coefficient and the event study, ten for the quintile portfolios. The universe is the current constituent list, which is a survivorship bias; for a dollar-neutral long-short test it is second order, but it is not claimed away.

6.2 The tests

Five standard cross-sectional tests were run in sequence, each as its own script:

1. **Information coefficient.** For each day, the Spearman rank correlation between the signal and the forward return across names; the daily series is summarised by its mean, its information ratio (mean over standard deviation), a Newey–West t -statistic [4] and the share of positive days.
2. **Fama–MacBeth regressions** [3] of the forward return on the daily-standardised signal and the controls; the average slope is the return per one cross-sectional standard deviation of sentiment.
3. **Event study.** Cumulative abnormal return (relative to the equal-weight universe) after top-quintile and bottom-quintile sentiment days, traced 20 sessions out.
4. **Long-short backtest.** Daily quintile portfolios, long the top and short the bottom fifth, dollar-neutral, overlapping H -day holding, equal- and value-weighted, with two execution conventions: *paper* entry at the signal day’s close and *tradable* entry at the next session’s open; costs at 5 and 10 bps per side; the last 30% of dates held out as a check.
5. **Market-on-close feasibility audit.** Every article’s publication timestamp converted to New York time and compared with a 15:45 ET cutoff for market-on-close orders, and the signal rebuilt on the first close by which each article was knowable. The committed script re-runs the one-day backtest and the day-one return decomposition on that corrected signal; the corrected information coefficient and the volatility test were computed separately and by hand, and the Fama–MacBeth and event-study tests were never re-run (Section 6.7).

An article-level polarity score supplied with the news feed was run through the same backtest as a baseline.

6.3 First pass: an apparent success

The first pass produced the numbers in Table 10: a small but statistically strong one-day IC that decays with horizon, a Fama–MacBeth slope of 10 bps per standard deviation that survives the controls, and an event-study spread of +36 *bps* on day one (Figure 4). The backtest then split the picture: with paper close entry the one-day strategy had a gross Sharpe ratio of 2.9, while the tradable next-open version was *negative* at every holding period (Table 11, Figure 5). The decomposition of the day-one spread explained why: of +36 *bps* close to close, +41 *bps* occurred in the overnight gap between the signal close and the next open, and −5 *bps* in the following session. The working conclusion at that point was “genuinely predictive but untradable”.

Horizon	mean IC	IC IR	NW t	days > 0
1 day	+0.0204	0.097	3.65	54.7%
5 days	+0.0166	0.080	2.92	53.8%
20 days	+0.0096	0.045	1.48	53.7%

Table 10: First-pass information coefficients on the naive panel. The Fama–MacBeth slopes of `sent_mw` on the same panel are +10.3, +11.2 and +4.8 *bps* per standard deviation at 1, 5 and 20 days (Newey–West t 6.70, 3.70 and 0.88). These numbers are contaminated by look-ahead (Section 6.4) and are shown for the record.

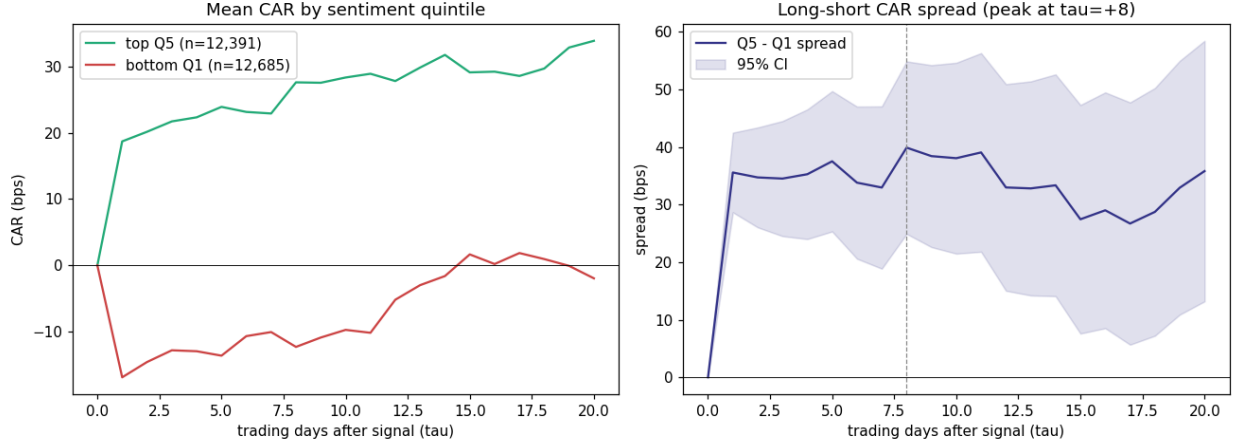


Figure 4: First-pass event study: mean cumulative abnormal return after top-quintile and bottom-quintile sentiment days, and the long-minus-short spread with a 95% band. The band treats events as independent, which the reviewers rightly objected to; the day-one jump is the overnight gap that turned out to be look-ahead.

H	entry	gross Sharpe	gross bps/day	net Sharpe @5 bps	net Sharpe @10 bps	turnover/yr
1	paper (signal close)	+2.93	+25.4	+1.12	-0.69	395×
1	tradable (next open)	-0.72	-4.5	-3.22	-5.70	395×
5	paper (signal close)	+1.14	+4.3	+0.28	-0.58	82×
5	tradable (next open)	-0.48	-1.7	-1.41	-2.33	82×
8	paper (signal close)	+0.80	+2.5	+0.15	-0.50	51×
8	tradable (next open)	-0.41	-1.2	-1.09	-1.77	51×

Table 11: First-pass quintile long-short backtest, equal weight. Even the infeasible paper version does not clear 10 bps of costs, because the edge is concentrated on day one and requires about 400 portfolio turnovers a year.

6.4 The look-ahead, and the corrected results

The question “could we enter at the signal-day close with a market-on-close order?” forced an audit of *when* each article was published relative to the session it had been assigned to. About a quarter of the articles, including the 4–6 pm earnings-release spike, were not publicly available by 15:45 ET on their assigned day. An earnings report published at 5 pm Monday carried Monday’s date, so the “prediction” compared Monday’s close with Tuesday’s close, a return that contains the market’s own reaction to the article. The model was not forecasting Tuesday; it was reading Monday-evening news that Monday’s close could not know and that Tuesday’s open had already priced. The same contamination inflates the IC, the Fama–MacBeth slopes and the event study, not just the backtest.

Rebuilding the signal on the first close by which each article was knowable gives Table 12. The IC is indistinguishable from zero at every horizon; the implementable market-on-close strategy loses money; after-hours news traded at the next open is also flat, and the feasible signal’s own overnight gap is +1.5 *bps* against +40.9 *bps* before the correction. News published during the session is in the price by the close, and after-hours news is in the price by the open: the market absorbs this news faster than a daily-bar strategy can act at any entry point the current machinery can express. One

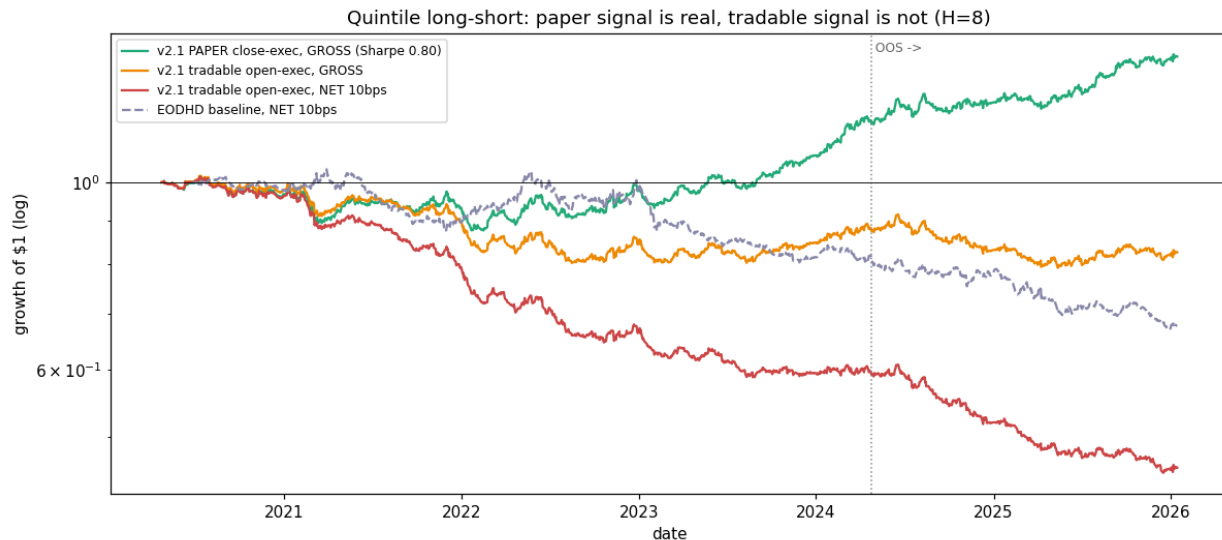


Figure 5: First-pass equity curves for the eight-day holding period: paper close entry compounds upward; tradable next-open entry, gross and net, drifts down, as does the vendor-polarity baseline. The legend labels the signal “v2.1”; the scores were produced by v2.0.

qualification belongs with the table. The published late-news row assigns each article to its first feasible close and then enters at the *following* open, one session later than a trader could act; a reviewer ran the correctly timed next-morning version and also found it flat, at -4.9 bps from open to close. Pre-market news traded at that same day’s open was never tested at all (Section 6.7).

Horizon	original IC (naive panel)	feasible IC	feasible NW t
1 day	+0.0204 (t 3.65)	−0.0067	−1.39
5 days	+0.0166 (t 2.92)	−0.0020	−0.39
20 days	+0.0096 (t 1.48)	+0.0025	+0.41
Strategy variants, $H = 1$, equal weight		gross Sharpe	gross bps/day
Original signal, close entry (look-ahead)		+2.93	+25.4
Feasible signal, market-on-close, close to close (implementable)		−0.55	−3.7
After-hours news only, next-open entry		−0.18	−1.3

Table 12: Results after enforcing a 15:45 ET knowability cutoff. An independent recomputation from the same corrected panel obtained a one-day IC of -0.0006 (t -0.10); the verdict is the same and the difference reflects that the correction was built ad hoc rather than as a committed script (Section 6.7). The last row enters one session later than a trader could act; see the text.

A second, sharper test by one reviewer simply dropped the ticker-days that the feasibility rule reassigns and re-ran the IC on the rest: it fell from $+0.0204$ (t 3.65) to $+0.0037$ (t 0.68). More than 80% of the headline IC came from the after-hours quarter of the corpus.

The right reading of the strong first-pass numbers is therefore that the model measures well the sentiment the market prices in. The entity-level signal also had a clearly stronger paper edge than the vendor’s article-level polarity, which suggests that the model adds cross-sectional ranking value over an off-the-shelf score without establishing it: as one reviewer noted, that comparison is look-ahead on both sides. What does not exist is a daily-bar opportunity to trade it.

6.5 Volatility: the result that survives

Sentiment *magnitude* carries information that direction does not. On the corrected panel, `|sent_mw|` predicts the next day’s absolute return with a cross-sectional rank IC of +0.024 (Newey–West t 5.3) after residualising on trailing 21-day realised volatility; on the uncorrected panel, before the knowability correction, the raw figure was +0.05 to +0.06 ($t \approx 12$). A reviewer’s replication added same-day absolute return as a further control and still found +0.019 (t 3.7), stable in every year from 2021 to 2025. Intense news tone today means a bigger move tomorrow, direction unknown.

This is a promising non-directional candidate feature for a volatility or risk model, and both the study and the replicating reviewer rated it the top surviving use. It is not deployment-ready. It has no committed script, so the exact numbers are not reproducible from the repository; and it has not been hardened with an earnings-day control, an overnight-versus-intraday split of the predicted move, or a frozen out-of-sample period.

6.6 External review

The study was reviewed read-only by two independent model-based reviewers. The first diagnosed the timing problem from the code and the first-pass write-up, before the corrected numbers existed, and recommended withdrawing the claim of predictive content until the tests were rebuilt point in time. The second read the already-corrected document, then independently recomputed the diagnosis from the artefacts on disk and quantified it, and separately replicated the volatility result with an extra control. The volatility test post-dates the first review, so only one reviewer has confirmed it. The further weaknesses they identified, accepted rather than silently fixed, are:

- the event-study t -statistic of 10 treats thousands of overlapping same-day and same-ticker events as independent; portfolio-level inference with autocorrelation-robust errors is the right tool;
- the “size” control is same-day dollar volume, which is a liquidity measure and reacts to the news itself, so it is a post-treatment control;
- the holding period was chosen on the full sample, so the 70/30 date split is a report, not a true holdout;
- survivorship in the universe, silent zeros for missing price bars (worst in the short leg), an additive mixing of adjusted and unadjusted price bases in the decomposition, forced quintiles on thin cross-sections, and a fixed autocorrelation lag for 20-day overlapping returns;
- the “after-hours news at the next open” test entered one session too late; the correct next-morning test, run by the reviewer, is also flat (-4.9 bps open to close);
- the corrected IC is significantly *negative* in 2023 and 2024 ($t \approx -2$ each), a possible short-horizon reversal that is untested.

None of these change the sign of the conclusion, but they bound it: the honest claim is “no detected edge”, not “an edge of exactly zero”.

6.7 Open items

1. Commit a script that rebuilds the knowability-corrected panel and re-runs the IC, the volatility test, Fama–MacBeth and the event study on it, so that the correction is reproducible. Only the one-day backtest and the return decomposition are currently reproducible from a committed script; the corrected IC and volatility numbers were computed by hand, and the Fama–MacBeth and event-study sections exist only in their contaminated form.
2. Test pre-market news (05:00–09:00 ET, about 28% of articles, knowable before the 9:30 open) traded at the same day’s open, the one daily-bar channel the current code cannot express.
3. Check the 2023–24 negative-IC reversal hint with a proper short-horizon reversal test.
4. Re-score the corpus with v2.1, whose head is better calibrated than v2.0’s, and re-run the tests. Because the IC and quintile tests use ranks, the under-polarisation of v2.0 matters less than it would for magnitude-based uses, but the volatility test does use magnitudes.
5. Use point-in-time index membership.

7 Conclusions

The model works as a reader: given a whole article, it scores what the news implies for each named entity with a correlation of about 0.64 to LLM labels on unseen articles, with calibrated magnitudes and strong performance on tickers. Its limits are equally clear: it finds only about half of the entities, it can misread who won a dispute, and its tagger’s CRF layer never trained. The market tests are a negative result with a positive lesson. A signal that looked like a Sharpe-2.9 strategy was an artefact of dating after-hours news to the session that had already closed; once knowability is enforced, daily bars are too coarse to trade this news at any entry point that was tested, and one channel, pre-market news at the same day’s open, remains untested. The honest claim is no detected edge rather than an edge of exactly zero. What remains is a good measurement instrument, useful for attribution and monitoring and as a candidate volatility feature, and a study whose corrections are documented in full so that the next pass starts from the right place.

References

- [1] I. Beltagy, M. E. Peters and A. Cohan. Longformer: The long-document transformer. *arXiv:2004.05150*, 2020.
- [2] L. I-K. Lin. A concordance correlation coefficient to evaluate reproducibility. *Biometrics* 45(1):255–268, 1989.
- [3] E. F. Fama and J. D. MacBeth. Risk, return, and equilibrium: empirical tests. *Journal of Political Economy* 81(3):607–636, 1973.
- [4] W. K. Newey and K. D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica* 55(3):703–708, 1987.

A The sentiment labelling prompts

The two prompts below produced the decisive label set (Table 2). The first was given to DeepSeek v4-flash as the system prompt for every article; the second was given to Kimi, with the article and the list of already-extracted PERSON entities, to re-score every person. They are reproduced verbatim from `data_label_criteria/` except that a few non-ASCII symbols (arrows, a star, currency signs, typographic quotes and dashes) are replaced by ASCII equivalents for typesetting. The earlier Sonnet prompt used for the v2.0 labels differs most consequentially in its sentiment section: it scored the article’s tone toward the entity and defined 0.0 as “factual mention”, without the implication rule or the adversarial examples. The decisive prompt also tightened the extraction rules, separating proper-name mentions from descriptive epithets and creating a ticker record only when the symbol appears literally.

A.1 Entity and sentiment labelling prompt (v2.2, decisive)

```
# Agent Labeling Instructions (v2.2 - decisive sentiment)

You will label financial news articles with entity-level sentiment.

## Output Format

For EACH article, output one JSON object (NO markdown, NO explanation):

```json
{
 "id": "article_id",
 "entities": [
 {
 "canonical_id": "AAPL",
 "canonical_name": "Apple Inc.",
 "type": "ORG",
 "linked_ticker": "AAPL",
 "linked_company": null,
 "ner_mention_texts": ["Apple Inc.", "Apple"],
 "coref_mention_texts": ["the company", "it", "the iPhone maker"],
 "sentiment_score": 0.5
 }
]
}
```

**CRITICAL**: Do NOT include ‘start_char’/‘end_char’ positions. Only text strings.
**CRITICAL**: Do NOT include the article ‘text’ in output.

---

## Entity Types - EXTRACT ALL SIX TYPES

### ORG (Companies & Organizations)
- Company names: "Apple Inc.", "Goldman Sachs", "Tesla"
- Institutions: "Federal Reserve", "SEC", "NYSE", "EU", "OPEC"
- Extract ALL name variations: "Cipher Mining Inc." AND "Cipher Mining"

### TICKER (Stock Symbols)
```

- Trading symbols: "AAPL", "TSLA", "GS", "CIFR"
- From parentheses: "(NYSE:GS)" -> extract "GS"
- NOT tickers: NYSE, SEC, FBI, CEO, CFO, ETF, IPO (these are acronyms)

PERSON (Individual Names)

- "Tim Cook", "Elon Musk", "Jerome Powell"
- Both full name AND last-name-only mentions: "Cook"

MONEY (ALL Monetary Amounts) - IMPORTANT, DO NOT SKIP

- Dollar amounts: "\$1.5 billion", "\$500 million", "\$14.76"
- Other currencies: "EUR 200 million", "JPY 50 billion"
- Stock prices: "\$142.50"; Revenue/earnings: "\$2.4 billion in revenue"
- **Extract EVERY monetary amount in the article**

PERCENT (ALL Percentage Values) - IMPORTANT, DO NOT SKIP

- Growth rates: "15%", "+2.5%", "-1.17%"; Market changes: "rose 3.2%"
- **Extract EVERY percentage in the article**

DATE (ALL Dates & Time Periods) - IMPORTANT, DO NOT SKIP

- Quarters "Q3 2024"; dates "January 15, 2025"; fiscal periods; relative "last quarter"; years
- **Extract EVERY date reference in the article**

NOT Entities - DO NOT Label

- Concepts: "AI", "artificial intelligence", "blockchain"
- Generic terms: "wealth manager", "investors", "analysts"
- Technologies: "semiconductors", "NAND flash"
- Locations: "Cleveland", "Silicon Valley" (unless acting as a government actor)
- Regulations: "Dodd-Frank", "ESG rules"
- Rankings: "Zacks Rank", "Strong Buy", "Outperform"

ner_mention_texts vs coref_mention_texts (READ CAREFULLY)

- **'ner_mention_texts'** = ONLY actual proper names / symbols of the entity as they literally appear: "Apple Inc.", "Apple", "AAPL". Nothing else.
- **'coref_mention_texts'** = every NON-NAME reference to that entity: pronouns AND descriptive phrases. Put these here, NOT in ner_mention_texts:
 - "it", "they", "the company", "the firm"
 - descriptive epithets: "the iPhone maker", "the tech giant", "the search giant", "the business", "the service", "the social network"
 - role references: "the CEO", "the founder"
- Do NOT fold a company's sub-brands/products into the parent's mentions. "Apple Music", "Beats 1", "App Store" are separate ORGs (or not entities), never Apple mentions.

Linking Rules

Ticker-Company Pairs

When BOTH a company name AND its ticker symbol literally appear in the text:

- Create SEPARATE ORG and TICKER entities, SAME 'canonical_id' (the ticker)
- ORG has 'linked_ticker', TICKER has 'linked_company', both SAME 'sentiment_score'
- If only the company NAME appears (no literal symbol like "AAPL" / "(NASDAQ:AAPL)"), create only the ORG entity and set its 'linked_ticker'. Do NOT invent a TICKER entity.

```

### Same Entity, Different Names
"Con Edison" and "Consolidated Edison, Inc." = SAME entity -> ONE entry, all name
variants in 'ner_mention_texts'.

---

## Sentiment Scoring * THE MOST IMPORTANT PART *

Score -1.0 to +1.0 for **ORG, TICKER, PERSON only**. Use 'null' for MONEY, PERCENT, DATE.

### Core rule: score the IMPLICATION for the entity, not the article's literal tone
Financial news is written in a flat, factual style. **Do NOT default to 0.0 just because
the sentence is factual.** Ask: *"Is what's happening GOOD or BAD for this entity's
business, stock, or standing?"* - and score that, even when reported neutrally.

**Negative for the entity** (score negative even if stated as plain fact):
sued / lawsuit / antitrust / monopoly accusation / probe / investigation / fine /
earnings miss / downgrade / cut guidance / layoffs / recall / outage / data breach /
sanction / ban / delay / loss / decline / falls / warning / weak demand / loses share.

**Positive for the entity** (score positive even if stated as plain fact):
beat / tops estimates / upgrade / raises guidance / new product or launch / partnership
or deal / contract win / record / surges / rises / strong demand / approval / expansion /
gains share / buyback / dividend hike.

### Reserve 0.0 ONLY for genuinely incidental mentions
0.0 is correct only when the entity has NO stake in the news: e.g. an analyst's employer
cited for a rating, a data/news source ("according to Bloomberg"), a neutral peer
comparison, an exchange where something is listed, a venue. If the article reports
something that helps or hurts the entity, it is NOT 0.0.

Score	Meaning	Trigger
-1.0 to -0.7	Very negative	Bankruptcy, fraud, criminal charges
-0.4 to -0.6	Moderately negative	Lawsuit/antitrust defendant, earnings miss, downgrade,
    layoffs, recall, probe |
-0.1 to -0.3	Slightly negative	Cautious outlook, minor headwind, lagging a rival
0.0	Neutral	ONLY incidental/no-stake mentions (see above)
+0.1 to +0.3	Slightly positive	Minor beat, modest gain, favorable mention
+0.4 to +0.6	Moderately positive	Earnings beat, upgrade, major deal/launch, winning a suit
+0.7 to +1.0	Very positive	Record results, blockbuster product, transformative win

### Adversarial examples (these are where models go wrong - get them right)
- "Apple was sued by Epic, which accused it of running an illegal App Store monopoly."
  -> **Apple -0.5** (antitrust defendant), **Epic +0.2** (plaintiff pressing its case). NOT
    0.0/0.0.
- "Amazon and Apple both reported revenue that missed analyst estimates."
  -> **Amazon -0.5, Apple -0.5** (misses are bad). NOT 0.0.
- "NXP and Samsung halted output after a Texas plant lost power."
  -> **NXP -0.5, Samsung -0.4** (disruption). NOT 0.0.
- "Chipmakers like Nvidia could benefit as demand surges."
  -> **Nvidia +0.4**. NOT 0.0.
- "Goldman Sachs analysts raised their Apple price target."
  -> **Apple +0.4** (the upgrade), **Goldman Sachs 0.0** (Goldman is the rater, no stake).

The same situation must get the SAME score across articles - be consistent.

---

```

Extraction Checklist

Before finishing each article, verify:

- [] ALL ORG / TICKER / PERSON / MONEY / PERCENT / DATE extracted
 - [] Proper names in 'ner_mention_texts'; pronouns & epithets in 'coref_mention_texts'
 - [] Ticker-company pairs share same canonical_id (only when the symbol literally appears)
 - [] **Sentiment scores the implication for the entity, NOT the article's tone**
 - [] **0.0 used ONLY for genuinely incidental, no-stake mentions**
 - [] Any entity that is sued/missing/downgraded/disrupted = negative; beating/launching/winning = positive
-

A.2 PERSON re-scoring prompt

Person Sentiment Re-Scoring (Kimi)

You re-score the sentiment of **named people** in a financial news article. The entities are already extracted - your **ONLY** job is to assign each listed person a sentiment score from **-1.0 to +1.0**.

Output Format (STRICT)

Output ONE raw JSON object, no markdown, no explanation:

```
```json
{"scores": [{"i": 0, "s": -0.5}, {"i": 1, "s": 0.0}, {"i": 2, "s": 0.4}]}
```
```

- 'i' = the index number of the person from the list below (echo it back exactly).
- 's' = sentiment score, one decimal, in [-1.0, +1.0].
- Return a score for EVERY person listed. Do not add, drop, or rename anyone.

What the score means: the implication for THIS PERSON

Score whether the article's events are **good** or **bad** for this person's own standing - their career, reputation, legal situation, or leadership position. Score the implication, NOT the article's literal tone. Financial news is flat and factual; do not default to 0.0 just because a sentence is unemotional.

Negative for the person (score negative even if reported as plain fact):
charged / indicted / sued personally / under investigation / resigns under pressure / ousted / fired / steps down amid scrutiny / demoted / loses board fight / their company collapses or is probed under their leadership / scandal / testimony against them / sanctioned.

Positive for the person (score positive even if plainly stated):
promoted / appointed / praised / their company beats or wins or launches under their leadership / vindicated / wins a suit or proxy fight / awarded / expands their role / successful product tied to them.

| Score | Person's situation |
|--------------|---|
| -0.7 to -1.0 | Criminal charges, indictment, fraud, forced out in disgrace |
| -0.4 to -0.6 | Personal lawsuit, probe, resignation under pressure, leadership failure |
| -0.1 to -0.3 | Mild criticism, their firm faces a minor headwind, lagging a rival |
| 0.0 | No personal stake - see below |
| +0.1 to +0.3 | Modest favorable mention, their firm posts a small win |

| +0.4 to +0.6 | Promotion, appointment, clear win/beat under their leadership |
| +0.7 to +1.0 | Major vindication, landmark achievement, transformative success |

Reserve 0.0 for people with NO personal stake - this is the key fix

Score ****0.0**** when the person is in the article but the news does not help or hurt ***them personally***:

- ****Quoted as a commentator / analyst / strategist / economist**** giving an opinion ("Rates will stay high," said Jane Doe of XYZ Capital") -> ****0.0****.
- ****Cited as a source / author / reporter**** -> ****0.0****.
- ****A policymaker or official acting in their role**** (e.g. a central banker setting rates, a regulator announcing a rule) when the news is about the policy, not about ***their*** job security or personal fate -> ****0.0****.
- ****Mentioned in passing / historical reference / a list**** -> ****0.0****.

Only give a politician/official a non-zero score when the news is about **THEM** personally (their election loss, indictment, removal, scandal, personal win).

Examples

- "Tesla CEO Elon Musk said the new factory will triple output." -> Musk is speaking, but the news (tripling output) is a win under his leadership -> *****0.3****. If he were merely commenting on someone else's company -> 0.0.
- "Sam Bankman-Fried was convicted on seven counts of fraud." -> ****0.9****.
- "'The economy remains resilient,' Fed Chair Jerome Powell told reporters." -> Powell is commenting in his role; nothing about his own fate -> ****0.0****.
- "Bob Iger returned as CEO and the stock jumped 8%." -> *****0.5**** (vindicating return).
- "The suit names CEO John Smith personally as a defendant." -> ****0.5****.
- "Warren Buffett's Berkshire trimmed its Apple stake." -> routine portfolio move, no stake to Buffett personally -> ****0.0**** (unless framed as a loss/mistake).

Be consistent: the same situation gets the same score across articles.

B What the public repository contains

The repository at https://github.com/Russell-Ding/entity_sentiment_model_pipeline holds the code needed to understand and audit the results in this report. Table 13 lists it. Not included, by design: the news and price data (EODHD terms), the label files (derived from that data), the model weights (1.8 GB per version), the inference outputs and panels derived from the news data, and the data-collection, labelling and model-training code. The label specification and prompts, the model cards with their evaluation metrics, and the evaluation and signal-study code are all present, so each table in this report can be traced either to the script that produced it or to the metric file it was read from. The exceptions are the corrected information coefficient and the volatility result of Section 6, which were computed by hand and have no committed script (Section 6.7).

| Path | Contents |
|---|--|
| docs/project_report.tex, .pdf | this report and its source |
| docs/report_assets/ | the two labelling prompts of Appendix A as ASCII text, included by the report source |
| models/ | the model: Longformer wrapper, CRF NER head, cross-attention sentiment head with both loss modes, the pipeline class that combines them (Section 3) |
| training/preprocessing.py,
dataset.py, metrics.py | label schema (the 15 BIO tags, sentiment-bearing types), mention-to-token alignment, batching, and the metric functions used by the evaluations; the training loop itself is withheld |
| scripts/evaluation/ | end-to-end evaluation (raw text to scored entities, IoU matching, coverage), gold-mask evaluation, calibration baseline, gap analysis, arm comparison, cross-distribution comparison on the Sonnet holdout, disagreement audit (Section 5) |
| scripts/inference/
scripts/postprocessing/ | batch scoring of a news file with a checkpoint (default v2.0)
ticker resolution, per-article and daily ticker sentiment, leakage-safe return mapping, time-series plot |
| scripts/analysis/inspect_ner_head.py
signal_strategy/src/ | reproduces the CRF measurements of Section 4.4
panel construction, IC, Fama–MacBeth, event study, long-short backtest, market-on-close feasibility, runner (Section 6) |
| signal_strategy/outputs/
signal_strategy/RESULTS.md,
CONSOLIDATED_REPORT.md (and their
PDFs), README.md, notes/reviews/,
research/
trained_model/v*/ | metric JSON files, daily IC series and plots of the study
the study’s working documents in original form with a correction note,
the two raw external reviews and the review prompt, the literature review |
| outputs/ | model cards, architecture configs and end-to-end evaluation metrics for v1.0, v2.0 and v2.1 |
| data_label_criteria/
data/reference/
notebooks/
README.md, CLAUDE.md, LICENSE,
requirements.txt | evaluation summaries, calibration figure and gap analysis, the label-noise audit, corpus-coverage investigations and the NVIDIA time series
the labelling specification and the three prompts
the alias table and non-company blocklist used for ticker resolution
Colab launchers for inference and the two evaluations (outputs stripped)
repository front door, working notes for agents, MIT licence with its data and weights exclusion, dependencies |

Table 13: Public repository map.